

Chien Nguyen

Portfolio Website | chiennguyen.uw.dev@gmail.com | LinkedIn: chien-nguyen03 | GitHub: chien-se

ABOUT

Computer Science student at UW Bothell specializing in cloud-native distributed systems, containerized microservices, and fault-tolerant backend infrastructure. Hands-on experience designing event-driven architectures with **Kafka, Kubernetes, Docker, and FastAPI**. Focused on high-availability system design and production reliability engineering.

EDUCATION

University of Washington Bothell

Expected Graduation: 2026

Bachelor of Science in Computer Science & Software Engineering

Bothell, WA

- **Relevant Coursework:** Parallel & Distributed Computing, Operating Systems, Databases, Computer Architecture, Data Structures & Algorithms, Machine Learning

Everett Community College

Graduated: September 2023

Associate in Arts and Sciences, Direct Transfer Agreement

Everett, WA

- Computer Science Honors Program (4.0 GPA)
- Vice President, Asian Student Union

PROJECTS

Real-Time Stock Analytics Pipeline | Kubernetes, Docker, Kafka, PostgreSQL

- Architected an event-driven, microservices pipeline ingesting **1,238+ streaming events/sec**, deploying services as isolated Kubernetes pods with horizontal autoscaling to maintain throughput under variable load.
- Reduced cascading failure risk across **5+ pipeline components** by implementing bulkhead isolation and health-check probes, achieving fault-tolerant recovery without full-system restarts.
- Instrumented all pipeline services with Prometheus metrics and Grafana dashboards, enabling visibility into Kafka consumer lag, pod health, and data throughput — mirroring production-grade observability practices.

Book Recommender System | Python, PySpark, Databricks (AWS EC2/S3)

- Engineered distributed ETL workflows across an AWS-hosted Spark cluster processing **20M+ records**, resolving executor instability by tuning partition strategies, shuffle behavior, and in-memory caching.
- Improved job completion reliability by **eliminating OOM-driven task failures** through adaptive query execution settings and Parquet/Delta storage format optimization.

Video Stream Platform Backend | SQL, FastAPI, PostgreSQL

- Designed a high-throughput session management backend processing **1,237+ concurrent sessions** across 254 devices, enforcing plan-aware concurrency limits through 4-stage transactional validation in an atomic operation.
- Reduced active-session query latency by **~40%** via 3 composite indexes targeting plan-subscription lookup patterns, supporting real-time anomaly detection at scale.

YouTube AI Model | Python, Ollama, REST APIs

- Built an AI-powered conversational agent using a custom **local LLM (Ollama)**, **reducing manual transcript extraction time 2×** by applying prompt engineering, context injection, and REST API integration.

SKILLS

Languages: Python, SQL, Java, C++

Data & ML: Pandas, NumPy, PyTorch, scikit-learn, Spark, Databricks (AWS), Hugging Face, Matplotlib

Systems & Tools: Apache Spark, Kafka, PostgreSQL, Docker, Kubernetes, FastAPI, Git, Prometheus, Grafana

Core Concepts: Distributed Systems, Scalable Systems, Cloud-native development, Agile Development, Microservices architecture